

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SentinelX — Cross-Camera Vehicle Intelligence & Unified CCTV Analytics Platform Gujarat Police Innovation Challenge 2026

Project Target:	Gujarat Police Innovation Challenge 2026
Document Version:	1.0.0 (Official Technical Release)
Cost & License:	■0 Licensing Cost · 100% Free & Open-Source (Apache-2.0 / MIT / BSD / PostgreSQL)
Architecture:	Hybrid Federated Edge-Metadata Intelligence (80,000 Camera Scalable)

1. Executive Summary & Purpose

SentinelX is a unified, vendor-neutral CCTV intelligence platform built for the Gujarat Police Innovation Challenge 2026. The system normalizes heterogeneous video streams across independent departmental VMS networks to deliver real-time vehicle tracking, license plate recognition (ANPR), journey reconstruction, and instant watchlist security alerts without commercial cloud or API dependencies.

Flagship User Story: Enter vehicle registration number -> Search indexed detections across cameras -> Reconstruct movement timeline -> Visualize route on OpenStreetMap GIS -> Check watchlists -> Dispatch real-time WebSocket Red Alerts -> Export SHA-256 verified evidence.

2. High-Level Architecture & Entity Schema

SentinelX combines dynamic catalog ingestion from /api/ingest, RTSP over TCP stream ingestion, AI vision workers (YOLOX + ByteTrack + PaddleOCR), and PostgreSQL/PostGIS metadata indexing.

Entity / Table	Key Attributes	Operational Function
cameras	id, external_camera_id, name, lat, lon, rtsp_url, whep_url, live_status	Dynamic CCTV registry consumed from /api/ingest
vehicle_events	id, camera_id, event_time, plate_normalized, vehicle_class, lat, lon	Searchable vehicle sighting metadata index
vehicle_tracks	id, camera_id, track_id, first_seen, last_seen, total_frames	ByteTrack single-camera multi-frame track continuity
vehicle_plates	id, event_id, plate_text, plate_normalized, confidence	PaddleOCR optical character readings and confidence
vehicle_embeddings	id, event_id, model_name, embedding_dim, vector_data	512-dim visual appearance vectors for Re-ID
watchlists	id, name, category (STOLEN, WANTED, SUSPICIOUS), is_active	Hotlists linked to active FIR cases
alerts	id, vehicle_event_id, status (NEW, ACKNOWLEDGED), alert_time	Real-time security hit alerts dispatched to dashboard
evidence	id, camera_id, file_path, sha256_hash, captured_at	Forensic snapshot vault with SHA-256 chain of custody
audit_logs	id, username, action, resource_type, timestamp, ip_address	Immutable append-only forensic user action trail

3. Functional Requirements (FR)

- **FR-01 (Dynamic Ingestion):** Query GET /api/ingest dynamically. Never hardcode camera counts or endpoints.
- **FR-02 (RTSP/TCP & Resilient Reconnection):** Enforce RTSP over TCP with exponential backoff (1s -> 30s) and bounded queues.
- **FR-03 (Detection & ByteTracking):** YOLOX detects vehicles; ByteTrack assigns persistent track IDs across frames.
- **FR-04 (ANPR OCR & Plate Normalization):** PaddleOCR extracts plates and normalizes registration formats (e.g. GJ01AB1234).
- **FR-05 (Historical Event Search):** Queries indexed metadata in < 200ms without seeking raw RTSP streams.
- **FR-06 (Cross-Camera Correlation Engine):** Reconstructs multi-camera journeys using travel-time plausibility filters.
- **FR-07 (Leaflet GIS Route Visualization):** Renders camera markers and journey polylines on interactive OpenStreetMap.
- **FR-08 (Watchlist & Real-Time Alerts):** Matches plates against watchlists and dispatches WebSocket Red Alerts in < 500ms.
- **FR-09 (Evidence Vault & SHA-256):** Records cryptographic SHA-256 hashes for all evidence snapshots.
- **FR-10 (RBAC & Forensic Audit Logging):** Enforces role access (Admin, Operator, Investigator) and logs all user actions.

4. 80,000-Camera Scalability & Evaluation Workflow

Statewide Scalability: SentinelX scales to 80,000 cameras using regional edge gateways that extract metadata locally, transmitting only lightweight JSON detection events (~1 KB) to the central command center, reducing network bandwidth by > 99%.

Mandatory Evaluation Script: 1. Dynamic Catalog Discovery -> 2. Live CCTV Grid -> 3. Vehicle Detection & ANPR -> 4. Vehicle Search (GJ01AB1234) -> 5. Cross-Camera Correlation -> 6. Movement Timeline & GIS Route -> 7. Watchlist Red Alert -> 8. Evidence & Audit.